

Question 1

BLAST® » blastp suite » results for RID-Z0E29HTZ01R

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Job Title 1336093|Genbank|Outer membrane integral membrane...
 RID Z0E29HTZ01R Search expires on 03-13 19:11 pm [Download All](#)
 Program BLASTP [Citation](#)
 Database nr [See details](#)
 Query ID lc|Query_617708
 Description 1336093|Genbank|Outer membrane integral membrane pr ...
 Molecule type amino acid
 Query Length 676
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Organism only top 20 will appear ☐ exclude
 Type common name, binomial, taxid or group name
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Percent Identity to E value to Query Coverage to
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Descriptions Graphic Summary Alignments Taxonomy

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Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1387	1387	100%	0.0	100.00%	676	WP_004155366.1

https://blast.ncbi.nlm.nih.gov/Blast.cgi?seqid=WP_012440288

Sequences producing significant alignments Download Select columns Show 100

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Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1387	1387	100%	0.0	100.00%	676	WP_004155366.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1385	1385	100%	0.0	99.85%	676	WP_168385176.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1385	1385	100%	0.0	99.85%	676	WP_168421624.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1371	1371	100%	0.0	98.97%	677	WP_004168436.1
☒ Type III secretion system outer membrane pore HrcC [Erwinia amylovora ATCC BAA-2158]	Erwinia amylo...	1369	1369	100%	0.0	98.82%	677	CBX79367.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia sp. Ejp617]	Erwinia sp. Ejp617	1347	1347	100%	0.0	98.89%	676	WP_014543268.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia pyrifoliae]	Erwinia pyrifoliae	1343	1343	100%	0.0	96.75%	676	WP_259816781.1
☒ HrcC [Erwinia pyrifoliae]	Erwinia pyrifoliae	1337	1337	100%	0.0	96.30%	676	ABA39798.2
☒ sulfatase-like hydrolase/transferase [Galdlineaceae bacterium]	Caldivineaceae b...	1278	1278	95%	0.0	96.58%	811	MG89158247.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia pirifloringrans]	Erwinia piriflori...	1269	1269	100%	0.0	92.46%	676	WP_023653761.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia tasmaniensis]	Erwinia tasmani...	1242	1242	97%	0.0	93.62%	676	WP_012440288.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia psidii]	Erwinia psidii	1211	1211	99%	0.0	86.67%	677	WP_124231871.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia tracheiphila]	Erwinia tracheip...	1184	1184	99%	0.0	83.85%	674	WP_233479868.1
☒ type III secretion system outer membrane ring subunit SctC [Pantoea]	Pantoea	1184	1184	100%	0.0	84.09%	679	WP_275222444.1
☒ type III secretion system outer membrane ring subunit SctC [Erwinia tracheiphila]	Erwinia tracheip...	1183	1183	99%	0.0	83.70%	674	WP_016191026.1
☒ type III secretion system outer membrane ring subunit SctC [Pantoea vagans]	Pantoea vagans	1180	1180	100%	0.0	83.95%	679	WP_061060948.1
☒ type III secretion system outer membrane ring subunit SctC [Pantoea agglomerans]	Pantoea agglom...	1176	1176	100%	0.0	83.65%	679	WP_098049835.1
☒ type III secretion system outer membrane ring subunit SctC [Pantoea agglomerans]	Pantoea agglom...	1176	1176	100%	0.0	83.80%	679	WP_323169453.1

Analysis

- All 100 similar sequences have an E value of 0.
- Out of these, 29 sequences show 100% query coverage.
- The minimum observed query coverage is 90%.
- Only one sequence has 100% identity, identified as the type III secretion system outer membrane ring subunit SctC in Erwinia amylovora.
- The lowest percentage identity recorded is 66.96%, attributed to the type III secretion system outer membrane ring subunit SctC in Dickeya lacustris.
- Therefore, it is likely that the provided sequence corresponds to the type III secretion system outer membrane ring subunit SctC in Erwinia amylovora.

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Job Title1336093(Genbank)Outer membrane integral membrane...
RIDZ0EX7UJH016Search expires on 03-13 19:25 pmDownload All
ProgramBLASTP
Citation
DatabaseSwissProt
See details
Query IDIcIQuery_7712450
Description1336093(Genbank)Outer membrane integral membrane pr...
Molecule typeamino acid
Query Length676
Other reportsDistance tree of resultsMultiple alignmentMSA viewer

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Organismonly top 20 will appear
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Type common name, binomial, taxid or group name

Add organism

Percent Identityto
E valueto
Query Coverage
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Sequences producing significant alignments
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☒ select all33 sequences selected
GenPeptGraphicsDistance tree of resultsMultiple alignmentMSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per Ident	Acc. Len	Accession
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Hypersensitivity response sec...	Pseudomonas sy...	566	566	98%	0.0	44.16%	701	Q01723.2
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=YscC secretin; Flags: Precurs...	Yersinia enterocol...	251	251	72%	2e-73	31.25%	607	Q01244.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Type III secretion protein Ysc...	Yersinia pestis	246	246	75%	1e-71	30.52%	607	Q56974.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Hypersensitivity response sec...	Ralstonia pseudo...	211	211	75%	5e-59	28.87%	568	Q52498.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Outer membrane protein Mxi...	Shigella flexneri	175	175	69%	6e-46	27.63%	566	Q04641.1
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Outer membrane protein Mxi...	Shigella sonnei	175	175	69%	6e-46	27.63%	566	Q55293.1
RecName: Full=SPI-2 type 3 secretion system secretin; Short=T3SS-2 secretin; AltName: Full=Outer membrane prot...	Salmonella enteri...	158	158	72%	2e-40	26.13%	497	D02WR9...
RecName: Full=Type 3 secretion system secretin; Short=T3SS secretin; AltName: Full=Hypersensitivity response sec...	Xanthomonas eu...	132	132	36%	5e-31	31.25%	607	P80151.1
RecName: Full=DNA utilization protein HoiQ; Flags: Precursor [Escherichia coli K-12]	Escherichia coli K...	104	104	37%	1e-22	30.00%	412	P34749.2
RecName: Full=Secretin ExsD; AltName: Full=General secretion pathway protein D; AltName: Full=Type II secretion...	Aeromonas hydro...	92.4	92.4	36%	3e-18	28.97%	678	P31780.2
RecName: Full=Secretin OutD; AltName: Full=General secretion pathway protein D; AltName: Full=Pectic enzymes s...	Dickeya chrysant...	92.0	92.0	41%	5e-18	28.05%	712	P31700.1
RecName: Full=Secretin ExsD; AltName: Full=General secretion pathway protein D; AltName: Full=Type II secretion...	Aeromonas salm...	90.1	90.1	36%	2e-17	28.42%	678	P45778.1
RecName: Full=Secretin OutD; AltName: Full=General secretion pathway protein D; AltName: Full=Pectic enzymes s...	Dickeya dadantii...	89.0	89.0	41%	4e-17	27.66%	710	Q01565.1
RecName: Full=Secretin OutD; AltName: Full=General secretion pathway protein D; AltName: Full=Pectic enzymes s...	Pectobacterium c...	86.7	86.7	84%	2e-16	22.80%	650	P31701.2
RecName: Full=Competence protein E; AltName: Full=DNA transformation protein ComE; Flags: Precursor [Haemop...	Haemophilus influ...	79.7	79.7	37%	2e-14	26.41%	445	P31772.2
RecName: Full=Putative secretin GscD; AltName: Full=Putative general secretion pathway protein D; AltName: Full=...	Escherichia coli K...	79.7	79.7	36%	3e-14	26.28%	650	P45758.2
RecName: Full=Secretin PulD; AltName: Full=General secretion pathway protein D; AltName: Full=Pullulanase secret...	Klebsiella pneum...	79.0	79.0	76%	6e-14	23.45%	660	P15644.1
RecName: Full=Embrall secretin protein DMO; Elasm: DMO secretin [Drosophila melanogaster] DMO11	Drosophila mel...	78.3	78.3	41%	4e-13	23.83%	714	P34750.3

Analysis

- All 33 similar sequences have E values close to 0.
- The Type 3 secretion system secretin in *Pseudomonas syringae* pv. *syringae* exhibits the highest query coverage at 98%.
- Three sequences share the lowest query coverage, recorded at 23%.
- The highest percentage identity is 44.16%, found in the Type 3 secretion system secretin in *Pseudomonas syringae* pv. *syringae*.
- The lowest percentage identity observed is 21.75%, associated with the Uncharacterized protein y4xJ in *Sinorhizobium fredii* NGR234.

Question 2

The following algorithm parameters were used for the previous question

blast.ncbi.nlm.nih.gov/blast.cgi?CLIENT=web&DATABASE=nr&NCBI_GI=on&PAGE=Proteins&PROGRAM=blastp&QUERY=IDQILETNRIACRFNHSNQKYAFSITFQEECAHVTLVVYGRNLHK...

☐ Show results in a new window

Algorithm parameters

[Restore default search parameters](#)

General Parameters

Max target sequences: 100
Select the maximum number of aligned sequences to display

Short queries: ☒ Automatically adjust parameters for short input sequences

Expect threshold: 0.05

Word size: 5

Max matches in a query range: 0

Scoring Parameters

Matrix: BLOSUM62

Gap Costs: Existence: 11 Extension: 1

Compositional adjustments: Conditional compositional score matrix adjustment

Filters and Masking

Filter: ☐ Low complexity regions

Mask: ☐ Mask for lookup table only
☐ Mask lower case letters

BLAST Search database nr using Blastp (protein-protein BLAST)
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Question 3

using the "Align two or more sequences" option in BLASTP.

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Align Sequences Protein BLAST

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Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

>1336093|Genbank|Outer membrane integral membrane protein|HrcC
MVEKRELRCRLGALLMLCATLPAGAQTADWKEQSYAYSADRTPLSTVLQ
DFADGHSVDLHLGNVEDTE
VTAKIRAEASAFILRLALEHHFQWVFYNNNTLYVSPQEQSSERLEISPDAA

Query subrange: From To

Or, upload file: No file chosen

Job Title: 1336093|Genbank|Outer membrane integral
Enter a descriptive title for your BLAST search

☒ Align two or more sequences

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

AAK81929.1

Subject subrange: From To

Or, upload file: No file chosen

Program Selection

Algorithm: ☒ blastp (protein-protein BLAST)
[Choose a BLAST algorithm](#)

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BLAST® » blastp suite-2sequences » results for RID-Z0FCEXJX114

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Job Title: 1336093|Genbank|Outer membrane integral
RID: Z0FCEXJX114 [Search expires on 03-13 19:33 pm](#) [Download All](#)

Program: Blast 2 sequences [Citation](#)

Query ID: Icl|Query_5888703 (amino acid)

Query Descr: 1336093|Genbank|Outer membrane integral membrane pr...

Query Length: 676

Subject ID: AAK81929.1 (amino acid)

Subject Descr: RscC [Pseudomonas fluorescens]

Subject Length: 713

Other reports: [Multiple alignment](#) [MSA viewer](#)

Filter Results

Percent Identity: to E value: to Query Coverage: to
[Filter](#) [Reset](#)

Descriptions

[Graphic Summary](#) [Alignments](#) [Dot Plot](#)

Sequences producing significant alignments

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☒ select all 1 sequences selected [GenPept](#) [Graphics](#) [Multiple alignment](#) [MSA Viewer](#)

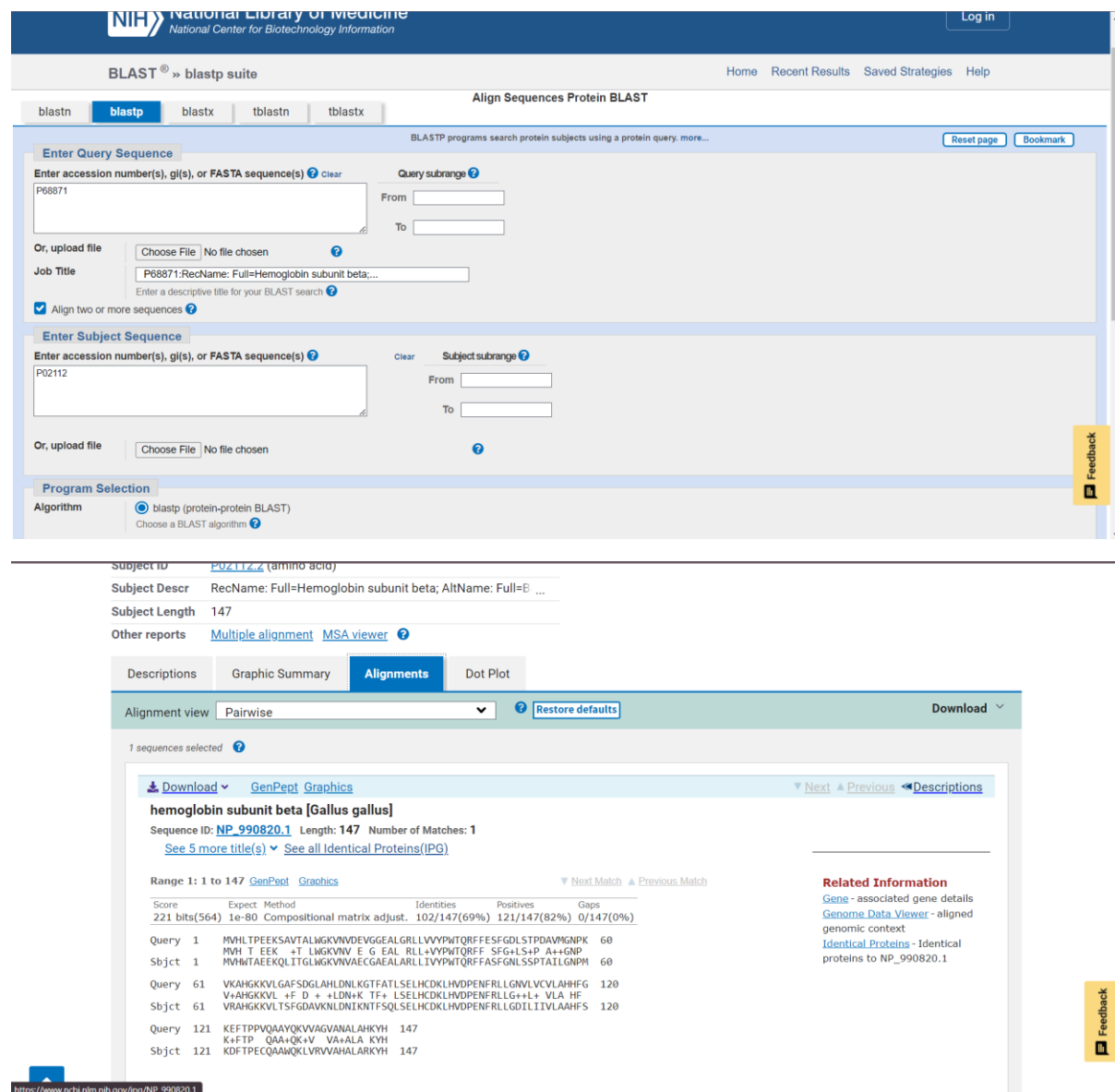
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per Ident	Acc. Len	Accession
☒ RscC [Pseudomonas fluorescens]	Pseudomonas fluorescens	530	530	96%	0.0	43.20%	713	AAK81929.1

[Feedback](#)

The percentage identity between the query sequence and AAK81929.1 was determined to be 43.20%.

Question 4

The alignment of human hemoglobin beta utilized the UniProtKB accession number P68871, while the alignment of chicken hemoglobin beta employed the UniProtKB accession number P02112. These accession numbers were utilized in conjunction with the "Align two or more sequences" option in BLASTP.



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Align Sequences Protein BLAST

BLASTP programs search protein subjects using a protein query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) [Query subrange](#)

P68871 From To

Or, upload file [Choose File](#) No file chosen

Job Title

P68871:RecName: Full=Hemoglobin subunit beta;... Enter a descriptive title for your BLAST search

☒ Align two or more sequences

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) [Subject subrange](#)

P02112 From To

Or, upload file [Choose File](#) No file chosen

Program Selection

Algorithm ☒ blastp (protein-protein BLAST) [Choose a BLAST algorithm](#)

Subject ID [P02112.4](#) (amino acid)

Subject Descr RecName: Full=Hemoglobin subunit beta; AltName: Full=B ...

Subject Length 147

Other reports [Multiple alignment](#) [MSA viewer](#)

Descriptions **Graphic Summary** **Alignments** **Dot Plot**

Alignment view [Pairwise](#) [Restore defaults](#) [Download](#)

1 sequences selected

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hemoglobin subunit beta [Gallus gallus]

Sequence ID: [NP_990820.1](#) Length: 147 Number of Matches: 1

[See 5 more title\(s\)](#) [See all Identical Proteins\(IPG\)](#)

Range 1: 1 to 147 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
221 bits(564)	1e-80	Compositional matrix adjust.	102/147(69%)	121/147(82%)	0/147(0%)

Query 1 MVHLTPPEKSAVTALWGKVNVDVEGGEALGRLLVYPNTQRFESFGDLSTPDAVGNPK 60
 MVH T EEK +T LWGKVN E G EAL RLL+VYPNTQRF SFG+LS+P A++GNP
 MVHWTAEKQLITGLWGKVNVAECGAELARLLIVYPNTQRFASFSGNLSPTAILGNPM 60

Sbjct 1 VRAHGKKVLTSFGDAVKNLNTIKNTFSQSELHCDKLHVDPENFRLLG+L+ VLA HF 120
 VRAHGKKVLTSFGDAVKNLNTIKNTFSQSELHCDKLHVDPENFRLLG+L+ VLA HF 120

Query 61 VRAHGKKVLTSFGDAVKNLNTIKNTFSQSELHCDKLHVDPENFRLLG+L+ VLA HF 120
 VRAHGKKVLTSFGDAVKNLNTIKNTFSQSELHCDKLHVDPENFRLLG+L+ VLA HF 120

Sbjct 61 VRAHGKKVLTSFGDAVKNLNTIKNTFSQSELHCDKLHVDPENFRLLG+L+ VLA HF 120
 VRAHGKKVLTSFGDAVKNLNTIKNTFSQSELHCDKLHVDPENFRLLG+L+ VLA HF 120

Query 121 KEFTPPVQAAYQKVVAGVANALAHKYH 147
 K+FTP QAA+QK+V VA+ALA KYH

Sbjct 121 KDFTECQAAMQKLVRRVAHALARKYH 147
 KDFTECQAAMQKLVRRVAHALARKYH 147

Related Information

[Gene](#) - associated gene details

[Genome Data Viewer](#) - aligned genomic context

[Identical Proteins](#) - Identical proteins to NP_990820.1

https://www.ncbi.nlm.nih.gov/ppg/NP_990820.1

The percentage similarity between human hemoglobin beta (P68871) and chicken hemoglobin beta (P02112) was determined to be 82%.

Question 5

```

In [1]: def find_pentapeptides(seq1, seq2):
    seq1_pentapeptides = {}
    for i in range(len(seq1) - 5 + 1):
        pentapeptide = seq1[i:i + 5]
        seq1_pentapeptides[pentapeptide] = seq1_pentapeptides.get(pentapeptide, 0) + 1

    matched_pentapeptides = {}
    for i in range(len(seq2) - 5 + 1):
        pentapeptide = seq2[i:i + 5]
        if pentapeptide in seq1_pentapeptides:
            matched_pentapeptides[pentapeptide] = matched_pentapeptides.get(pentapeptide, 0) + 1

    for pentapeptide, frequency in matched_pentapeptides.items():
        freq_seq1 = seq1_pentapeptides[pentapeptide]
        freq_seq2 = frequency - freq_seq1
        print(f"{pentapeptide} occurs {frequency} time(s) in both peptides "
              f"({freq_seq1} time(s) in Seq. 1 and {freq_seq2} time(s) in Seq. 2)")

human_sequence = "MMHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMVGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFTLSELHCDKLHVDPENFRLLGNVLI
chicken_sequence = "MMHWTAEEKQLITGLWGKVNVAECGAELARLLIVYPWTQRFFASFGNLSSTAILGNPMVRAHGKKVLTSFGDAVKNLNDIKNTFSQLSELHCDKLHVDPENFRLLG

find_pentapeptides(human_sequence, chicken_sequence)

```

```

LWGKV occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
WGKVN occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
GKVVN occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
VYPWT occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
YPTQ occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
PWTQR occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
WTQRF occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
TQRF occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
AHGKK occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
HGKKV occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
GKKVL occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
LSELH occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
SELHC occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
ELHCD occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
LHCDK occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
HCDKL occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
CDKLH occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
DKLVH occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
KLHVD occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
LVHVD occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
HVDPE occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
VDPEN occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
DPENF occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
PENFR occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
ENFRL occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
NFRLL occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)
FRLLG occurs 1 time(s) in both peptides (1 time(s) in Seq. 1 and 0 time(s) in Seq. 2)

```

Question 6

```

In [5]: def load_blosum62():
        # Define the BLOSUM62 substitution matrix manually
        blosum62 = {
            ('A', 'A'): 4, ('A', 'R'): -1, ('A', 'N'): -2, ('A', 'D'): -2, ('A', 'C'): 0,
            ('A', 'Q'): -1, ('A', 'E'): -1, ('A', 'G'): 0, ('A', 'H'): -2, ('A', 'I'): -1,
            ('A', 'L'): -1, ('A', 'K'): -1, ('A', 'M'): -1, ('A', 'F'): -2, ('A', 'P'): -1,
            ('A', 'S'): 1, ('A', 'T'): 0, ('A', 'W'): -3, ('A', 'Y'): -2, ('A', 'V'): 0,
            # ... (Complete BLOSUM62 matrix)
        }
        return blosum62

def calculate_scores(seq1, seq2):
    # Use the BLOSUM62 matrix
    matrix = load_blosum62()

    # Perform global alignment
    alignments = align_sequences(seq1, seq2, matrix, -10, -0.5)

    # Get the best alignment (first one in the list)
    best_alignment = alignments[0]

    # Calculate identity score
    matches = sum(a == b for a, b in zip(best_alignment[0], best_alignment[1]))
    total_length = max(len(seq1), len(seq2))
    identity_score = (matches / total_length) * 100

    # Calculate similarity score
    similar = sum(matrix.get((a, b), 0) > 0 for a, b in zip(best_alignment[0], best_alignment[1]))
    similarity_score = (similar / total_length) * 100

    # Calculate gap percentage
    gap_count = best_alignment[0].count('-') + best_alignment[1].count('-')
    gap_percentage = (gap_count / total_length) * 100

    # Calculate query coverage score
    query_coverage = (len(seq2) / len(seq1)) * 100

    return identity_score, similarity_score, gap_percentage, query_coverage

def align_sequences(seq1, seq2, matrix, gap_open, gap_extend):
    # Perform global alignment without using pairwise2 module
    # (Implementation of Needleman-Wunsch algorithm)
    m, n = len(seq1), len(seq2)

    # Initialize the scoring matrix
    score_matrix = [[0] * (n + 1) for _ in range(m + 1)]

    # Fill in the scoring matrix
    for i in range(1, m + 1):
        for j in range(1, n + 1):
            match = score_matrix[i - 1][j - 1] + matrix.get((seq1[i - 1], seq2[j - 1]), -10)
            delete = score_matrix[i - 1][j] - gap_extend
            insert = score_matrix[i][j - 1] - gap_extend
            score_matrix[i][j] = max(match, delete, insert)

    # Traceback to find the alignment
    aligned_seq1, aligned_seq2 = '', ''
    i, j = m, n
    while i > 0 or j > 0:
        if i > 0 and j > 0 and score_matrix[i][j] == score_matrix[i - 1][j - 1] + matrix.get((seq1[i - 1], seq2[j - 1]), -10):
            aligned_seq1 = seq1[i - 1] + aligned_seq1
            aligned_seq2 = seq2[j - 1] + aligned_seq2
            i -= 1
            j -= 1
        elif i > 0 and score_matrix[i][j] == score_matrix[i - 1][j] - gap_extend:
            aligned_seq1 = seq1[i - 1] + aligned_seq1
            aligned_seq2 = '-' + aligned_seq2
            i -= 1
        else:
            aligned_seq1 = '-' + aligned_seq1
            aligned_seq2 = seq2[j - 1] + aligned_seq2
            j -= 1

    return [(aligned_seq1, aligned_seq2)]

# Test with two protein sequences
seq1 = "MVHLTPKSAVTALWGKVNVDVGEALGRLLVVPWTRQFFESFGDLSTPDVVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLVHPDENFRLLGNVLVCVLAHHFG"
seq2 = "MVHNTAEKOLITGLWGKVNVAECGAEALARLLIVYPWTRFFASFGNLSPTAILGNPMVRAHGKKVLTSFGDAVKNLONIKNTFSOLSELHCDKLVHPDENFRLLGDILLIIVLAHIFS"

```



```

for i in range(1, m + 1):
    for j in range(1, n + 1):
        match = score_matrix[i - 1][j - 1] + matrix.get((seq1[i - 1], seq2[j - 1]), -10)
        delete = score_matrix[i - 1][j] - gap_extend
        insert = score_matrix[i][j - 1] - gap_extend
        score_matrix[i][j] = max(match, delete, insert)

# Traceback to find the alignment
aligned_seq1, aligned_seq2 = '', ''
i, j = m, n
while i > 0 or j > 0:
    if i > 0 and j > 0 and score_matrix[i][j] == score_matrix[i - 1][j - 1] + matrix.get((seq1[i - 1], seq2[j - 1]), -10):
        aligned_seq1 = seq1[i - 1] + aligned_seq1
        aligned_seq2 = seq2[j - 1] + aligned_seq2
        i -= 1
        j -= 1
    elif i > 0 and score_matrix[i][j] == score_matrix[i - 1][j] - gap_extend:
        aligned_seq1 = seq1[i - 1] + aligned_seq1
        aligned_seq2 = '-' + aligned_seq2
        i -= 1
    else:
        aligned_seq1 = '-' + aligned_seq1
        aligned_seq2 = seq2[j - 1] + aligned_seq2
        j -= 1

return [(aligned_seq1, aligned_seq2)]

# Test with two protein sequences
seq1 = "MVHLTPEEKSAVTALGKVNVDVGVGGEALGRLLVVYPWTRQFFSGDLSPTDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGFTATLSELCDDKLVHDPENFRLLGNNVLCVLAHHFG
seq2 = "MVHMTAEKQLITGLGKVNVAECGAELARLLIVYPWTRQFFASFGNLSPTAILGNPMVRAHGKKVLTSGDAVKNLNDIKNTFSQELSELCDDKLVHDPENFRLGDIILVLAHHFG

identity_score, similarity_score, gap_percentage, query_coverage = calculate_scores(seq1, seq2)

print(f"Identity Score: {identity_score}")
print(f"Similarity Score: {similarity_score}")
print(f"Gap Percentage: {gap_percentage}%")
print(f"Query Coverage: {query_coverage}%")

```

Identity Score: 10.204081632653061
 Similarity Score: 10.204081632653061
 Gap Percentage: 179.59183673469389%
 Query Coverage: 100.0%

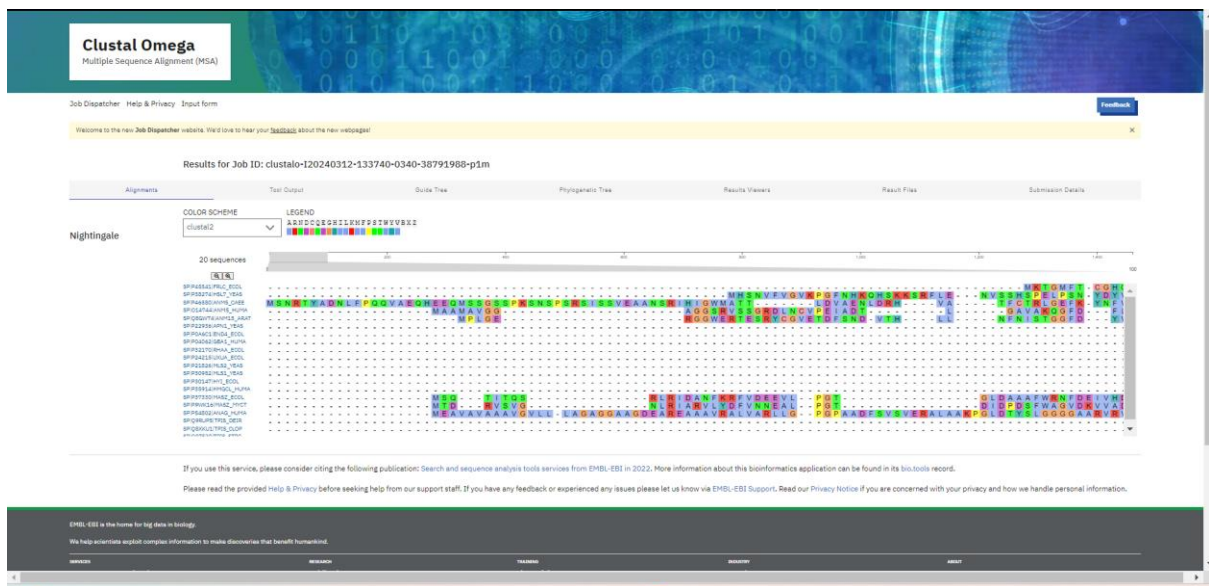
Question 7

UniProtKB 500,408 results

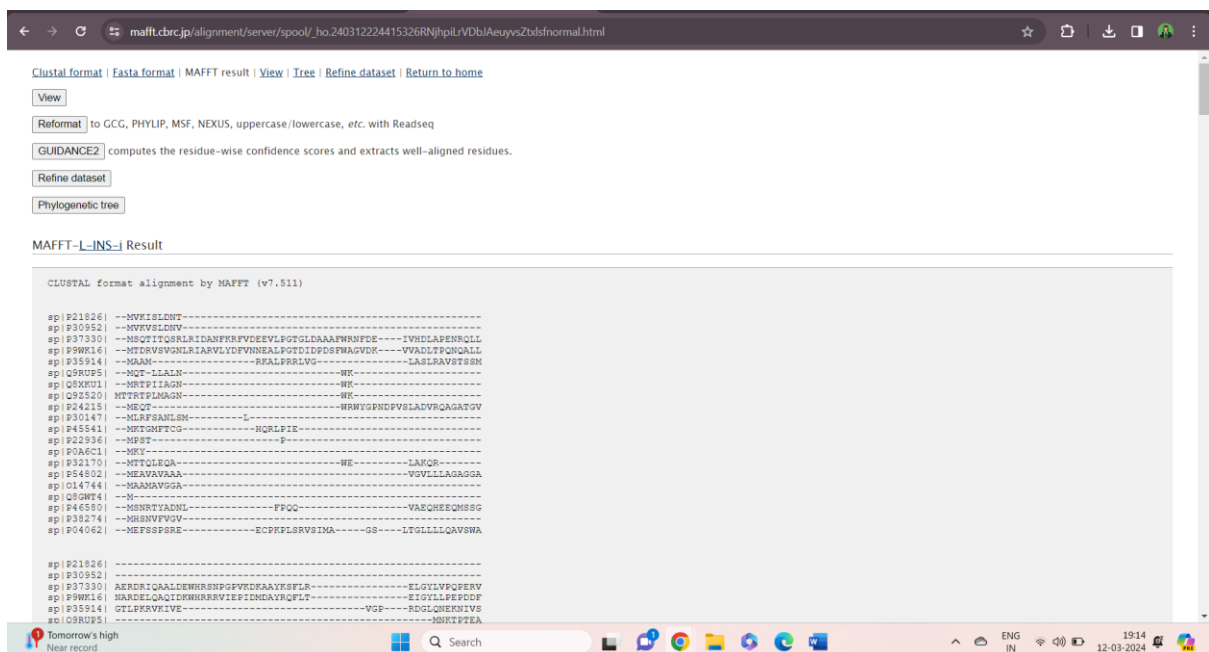
Entry	Entry Name	Protein Name	Gene Name	Organism	Length
P21826	MLS2_YEAST	Malate synthase 2[...]	DAL7, MLS2, YIR031C	Saccharomyces cerevisiae (strain ATCC 204508 / S288c) (Baker's yeast)	554 AA
P37330	MASZ_ECOLI	Malate synthase G[...]	glcB, glc, b2976, JW2943	Escherichia coli (strain K12)	723 AA
O14744	ANM5_HUMAN	Protein arginine N-methyltransferase 5[...]	PRMT5, HRMT1L5, IBP72, JBP1, SKB1	Homo sapiens (Human)	637 AA
P54802	ANAG_HUMAN	Alpha-N-acetylglucosaminidase[...]	NAGLU, UFHSD1	Homo sapiens (Human)	743 AA
P35914	HMGCL_HUMAN	Hydroxymethylglutaryl-CoA lyase, mitochondrial[...]	HMGCL	Homo sapiens (Human)	325 AA
P30952	MLS1_YEAST	Malate synthase 1[...]	MLS1, YNL117W, N1921	Saccharomyces cerevisiae (strain ATCC 204508 / S288c) (Baker's yeast)	554 AA
P24215	UXUA_ECOLI	Mannoside dehydratase[...]	uxuA, b4322, JW4285	Escherichia coli (strain K12)	394 AA
P22936	APN1_YEAST	Apurinic-apyrimidinic endonuclease 1[...]	APN1, YKL114C, YKL513	Saccharomyces cerevisiae (strain ATCC 204508 / S288c) (Baker's yeast)	367 AA
P45541	FRLC_ECOLI	Fructoseylsine 3-epimerase[...]	frlC, yfH, yfR, b4474, JW5699	Escherichia coli (strain K12)	276 AA
P30147	HYL_ECOLI	Hydroxypyruvate isomerase[...]	hyl, glp, ybbG, b0508, JW0496	Escherichia coli (strain K12)	258 AA
P0A6C1	END4_ECOLI	Endonuclease 4[...]	nfo, b2159, JW2146	Escherichia coli (strain K12)	285 AA
Q9RUP5	TPIS_DEIRA	Triosephosphate isomerase[...]	tpiA, DR_1339	Deinococcus radiodurans (strain ATCC 13939 / DSM 20539 / JCM 16871 / CCUG 27074 / LMG 4051 / NBRC 15346 / NCIMB 9279 / VKM B-1422 / R1)	244 AA
P04062	GBA1_HUMAN	Lysosomal acid glucosylceramidase[...]	GBA1, GBA, GC, GLUC	Homo sapiens (Human)	536 AA
P46580	ANM5_CAEEL	Protein arginine N-methyltransferase 5[...]	prmt-5, tag-251, C34E10.5	Caenorhabditis elegans	734 AA
Q8XKU1	TPIS_CLOPE	Triosephosphate isomerase[...]	tpiA, CPE1302	Clostridium perfringens (strain 13 / Type A)	248 AA
P32170	RHAA_ECOLI	L-rhamnose isomerase[...]	rhaA, b3903, JW5561	Escherichia coli (strain K12)	419 AA
Q9Z520	TPIS_STRCO	Triosephosphate isomerase[...]	tpiA, tpi, SCO1945, SCC5405c	Streptomyces coelicolor (strain ATCC BAA-471 / A3(2) / M145)	258 AA
P9WK16	MASZ_MYCTO	Malate synthase G[...]	glcB, MT1885	Mycobacterium tuberculosis (strain CDC 1551 / Oshkosh)	741 AA
P38274	HSL7_YEAST	Protein arginine N-methyltransferase HSL7[...]	HSL7, YBR133C, YBR100B	Saccharomyces cerevisiae (strain ATCC 204508 / S288c) (Baker's yeast)	

The twenty sequences were aligned using Clustal Omega, MAFFT, and MUSCLE:

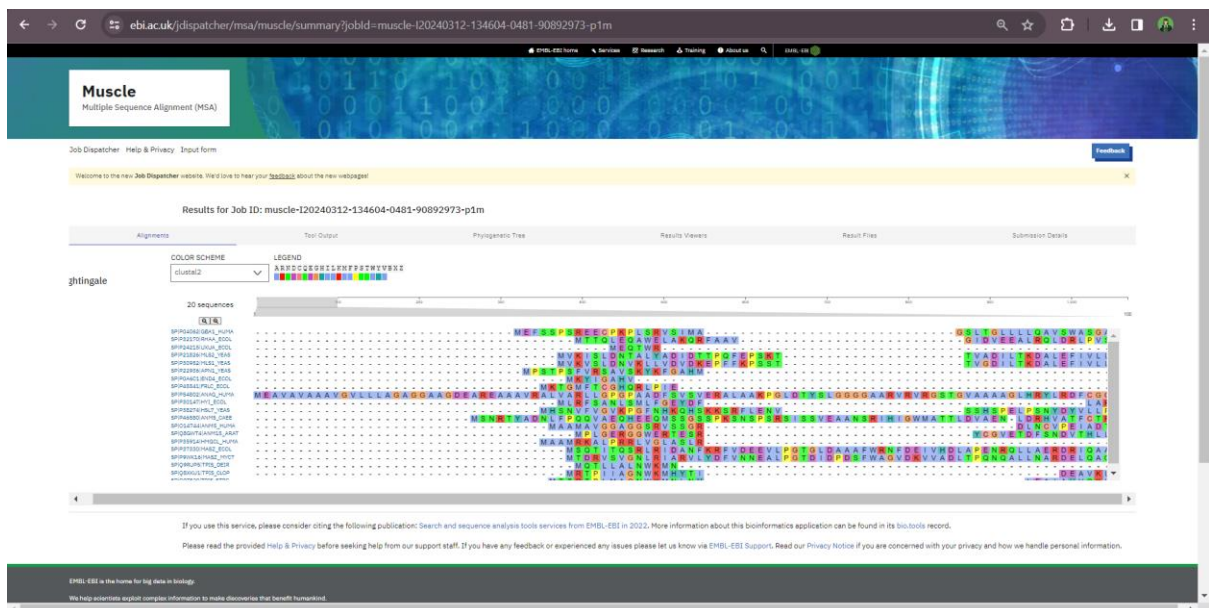
Clustal Omega



MAFFT



MUSCLE



The five residue positions that are aligned differently in the three methods are

- SP|P45541|FRLC_ECOL

- SP|P38274|HSL7_YEAS
- SP|P46580|ANM5_CAEE
- SP|O14744|ANM5_HUMA
- SP|Q8GWT4|ANM15_ARAT

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Description	Scientific Name	Max	Total	Query	E	Per.	Acc.
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Escherichia coli str. K-12 substr. W3110]	Escherichia coli...	53.2	53.2	100%	2e-10	100.00%	420
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Escherichia coli K-12]	Escherichia coli...	53.2	53.2	100%	2e-10	100.00%	410
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Escherichia coli K-12]	Escherichia coli...	53.2	53.2	100%	2e-10	100.00%	406
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Escherichia coli]	Escherichia coli...	53.2	53.2	100%	2e-10	100.00%	398
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Escherichia coli K-12]	Escherichia coli...	53.2	53.2	100%	2e-10	100.00%	398
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Escherichia coli]	Escherichia coli...	53.2	53.2	100%	2e-10	100.00%	398
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Escherichia coli]	Escherichia coli...	43.1	43.1	100%	8e-07	85.71%	406
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Escherichia coli]	Escherichia coli...	43.1	43.1	100%	8e-07	85.71%	400
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Zymomonas mobilis]	Zymomonas mob...	42.2	42.2	92%	2e-06	84.62%	388
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Yersinia pseudotuberculosis YPIII]	Yersinia pseudot...	40.9	40.9	85%	5e-06	91.67%	401
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Acinetobacter baumannii AB307-0294]	Acinetobacter ba...	35.4	35.4	92%	4e-04	76.92%	406
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Moraxella catarrhalis]	Moraxella catarrh...	33.3	33.3	85%	0.002	75.00%	432
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Moraxella catarrhalis]	Moraxella catarrh...	33.3	33.3	85%	0.002	75.00%	415
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Vibrio vulnificus CMCP6]	Vibrio vulnificus...	28.2	28.2	85%	0.15	75.00%	427
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Vibrio vulnificus CMCP6]	Vibrio vulnificus...	28.2	28.2	85%	0.15	75.00%	405
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase 2 [Arabidopsis thaliana]	Arabidopsis thali...	26.1	39.0	92%	0.87	69.23%	414
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Streptomyces ficus]	Streptomyces fic...	24.4	24.4	42%	3.5	100.00%	384
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase 1 [Arabidopsis thaliana]	Arabidopsis thali...	24.4	37.3	92%	3.5	64.29%	414
Chain Bb_30S ribosomal protein S9_mitochondrial [Arabidopsis thaliana]	Arabidopsis thali...	24.4	24.4	42%	3.5	100.00%	430
Chain A_1-deoxy-D-xylulose 5-phosphate reductoisomerase [Thermotoga maritima]	Thermotoga mari...	24.0	24.0	50%	5.0	85.71%	376
Chain B_Neural acetylcholine receptor subunit beta-2 [Homo sapiens]	Homo sapiens	24.0	24.0	71%	5.0	63.64%	397
Chain B_Neural acetylcholine receptor subunit beta-2 [Homo sapiens]	Homo sapiens	24.0	24.0	71%	5.0	63.64%	403
Chain B_Neural acetylcholine receptor subunit beta-2 [Homo sapiens]	Homo sapiens	24.0	24.0	71%	5.0	63.64%	487
Chain A_Tricorn protease-interacting factor F3 [Thermoplasma acidophilum DSM 1728]	Thermoplasma a...	24.0	24.0	64%	5.0	66.67%	780

- Many results show 100% query coverage, high percentage identity, and E values close to 0.
- The query sequence closely resembles subsequences found in Chain A of 1-deoxy-D-xylulose 5-phosphate reductoisomerase in Escherichia coli str. K-12.
- The graphic summary indicates a strong relationship between the provided query sequence and the Dxr superfamily.

